

# EEF Meeting Minutes, Thursday Jan 21, 2021

Erlang Ecosystem Foundation, Minutes of a Regular Meeting of the Board of Directors Meeting #41, Jan 21, 2021

A regular meeting of the Erlang Ecosystem Foundation was held over group conference call (Zoom). The meeting began at 16:10 UTC.

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## Attendance

The following members of the board of directors were present (10/11): Richard Carlsson, Francesco Cesarini, Benoît Chesneau, Maxim Fedorov, Kenneth Lundin, Peer Stritzinger, Sebastian Strollo, José Valim, Miriam Pena Villanueva, and Alistair Woodman.

Bryan Paxton was also present

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## Agenda Items

1. Vote of the minutes of the previous meeting.

*RESOLVED, that the Erlang Ecosystem Foundation approve the minutes as presented as representing a true and accurate record of the board meeting #40 held on Dec 17, 2020. Unanimously approved.*

2. Intermediate actions: marketing contract still being worked on, Alistair hoping to wrap up soon.
3. Treasury update: We have approximately \$146,000 in the bank. Sebastian will send 1099 to IRS, and tax receipts to all members and sponsors who gave us money in 2020. Membership update: pretty much unchanged.
4. Alistair suggested that he looks into clarifying section 5.12 in our bylaws to streamline and facilitate online voting for the board when applicable. None against.
5. Prepared topics that needed a board vote:
  1. Request from Benoît to have permission to sign up for GSOC 2021 (to suggest projects, see <https://github.com/erlef/gsoc>)

*RESOLVED, that the EEF approve Benoît's request to sign up for Google Summer of Code 2021. Unanimously approved.*

2. Stipend request to support development of JIT for the ARM family of processors (see attached stipend request).

*RESOLVED, that the EEF approve supporting the development of JIT for the ARM family of processors up to \$24,000. Unanimously approved.*

3. Stipend request to support development of improved float to string conversion in OTP (see attached stipend request).

*RESOLVED, that the EEF approve supporting the development of the improved float to string conversion up to \$3,000. Unanimously approved.*

4. Proposal from the fellowship working group.

*RESOLVED, that the EEF approve the fellowship group to proceed with the following proposals: "Remove the fellows from the bylaws through board approval.", "Work with the community to agree and propose new service awards.", and "If the work group feels there is a need to give a lifetime voting membership to certain fellows, they will request funding from the board like any other work group or sponsorship from a generous donor.". Vote count: For: 9, Against: 0, Abstain: 1.*

6. Demo from Bryan of the latest updates to the EEF site. All in all great progress!

- A calendar has been added, which contains all the EEF board and working group meetings (plan to add direct links to meetings etc)
- Tools have been added for the working group chairs to add members and chairs, edit / view / submit reports (submitted to board report private github repo)
- Administrative interfaces to add sponsors, and to edit details of working groups

7. Discussion around the status of the marketing working group. It has been without a chair for some time, and without much progress. Alistair suggested that with the contract in place to have someone that can act on things that need to be done the members in the wg will become more active. Benoît raised a concern that it may give too much power to an external company.

8. Alistair declared the meeting adjourned at 17:07 UTC.

Dated Jan 21, 2021

Secretary of the meeting: \_\_\_\_\_ Sebastian Strollo

Attachments:

1. Stipend request to support development of JIT for the ARM family of processors (3 pages)
2. Stipend request to support development of improved float to string conversion in OTP (2 pages)

## General info

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## Proposal

Sherpa: José Valim  
Last version: 13/01/2021

## Bio

I'm an independent software engineer that offers a commercial support for [AsmJit](#) and [Blend2D](#) libraries in addition to consultations and contracted work on any other project, be it open source or proprietary. I prefer to work on projects where performance matters and where C++ language is used for its efficiency and not for legacy reasons. I can use other programming languages too, like Python or JavaScript/Node.js, but I'm not actively looking for opportunities that would involve these languages without the use of C++.

## Purpose

Erlang/OTP 24 will include a JIT compiler. It is built on top of AsmJit, which currently supports only x86 archs. While ARM architectures have always been used within the Erlang Ecosystem, especially in the embedded camp, Apple's M1 chip and the latest Raspberry PI 4 bring growing attention and interest to ARM64.

The goal of this proposal is to sponsor 480 hours of work on AsmJit, in order to bring ARM64 and ARM32 support. The work will be part-time, which means 20 hours a week, ongoing for 24 weeks. This allows plenty of time to think about different approaches to the problems that need to be addressed.

I will charge USD50/hour which brings the project to a budget of USD24000 spread over several months. I would appreciate it if work can start from February 1st.

## Milestones

There are two major milestones, which are deliverables, which I break down below:

- AArch64 (USD15000) - The goal is to support all instructions except SVE (this instruction set is only used by Fujitsu supercomputers, not provided by Apple M1). In particular:
  - Fix asmjit to work with Apple M1 CPU on MacOS. There is already a PR that tries to fix something, but it's incomplete and I will have to do some work in this area. There are some new APIs that Apple provides for a hardened runtime (which is enforced), so some changes will be necessary in how asmjit works with executable memory (VirtMem and JitAllocator APIs, not a big deal).
  - Introduce a real database of AArch64 instructions that we can generate the instruction tables from (including the generator). This should make it easy to introduce new instructions and to provide more features like instruction validation and query.
- ARM32 (USD9000) - Basic instruction set only (and no Thumb initially). At the moment, it is hard to fully estimate, but it would be nice to have infrastructure around ARM32 working (like branches, relocations, etc) with the possibility to add more instructions later.

If by any chance there is extra time, it will be used to improve AsmJit in general. I have the following ideas:

- Add new AArch64 instruction sets, like SVE and SVE2 (not sure this is hot at the moment, there is most likely only one server-grade CPU that supports these)
- Work on ARM/Thumb
- Not sure anything else has a possibility to be worked on during this time

In general it's a lot of work for 3 full months. I understand the priority of 64-bit ARM targets so I would consider spending more time on AArch64. This means that I'm not sure there will be a fully usable 32-bit ARM backend after the contract is over but I will try my best and at least provide basic ISA support.

## Reports

I will send an invoice at the end of every month alongside a brief report containing:

- A daily break-down of how many hours have been invested into the project for that given month for invoice purposes
- 1 or 2 paragraphs with a summary of the work done and what is next

The Erlang Ecosystem Foundation will also be listed as a donator to the asmjit project.

## Sherpa notes

The Erlang/OTP team has already manifested interest in the work and agrees the changes will be beneficial to the Erlang VM. Both Build and Packaging and Embedded WG have manifested interest in moving this proposal forward.

# Speeding up float to string conversion

## General info

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## Proposal

Sherpa: Peer Stritzinger

Last version: <date>

## Bio

29, french Thomas is from France. He sees himself both as a Product Oriented Engineer and a Product Minded Engineer . He holds a bachelor in Science and Technologies, with a specialty in CS. He is working presently as a Freelance Developer remotely from France. He spent the past few years helping teams run faster and focus on what matters, through product focus and use of Elixir. He has a keen interest in Cognitive Systems Engineering. His personal interests range from spaceships to Human Factors, but talking about it would take a serious amount of time and a reasonable amount of tea. He has a particular interest in building tooling, back-end technologies and bringing products to the next level.

## Purpose

Speed up float to string conversion on the BEAM by implementing the Ryu algorithm. First, by replacing the `io_lib_format:fwrite_g/1` function implementation to use Ryu, purely in erlang.

Then by offering an EEP to replace the BIF for `erlang:float_to_list/2` and `erlang:float_to_binary/2` implementation, from using the `libc sprintf` into an erts implementation of Ryu, and working on merging `io_lib_format:fwrite_g/1` into the BIF as a new formatting options ("shortest").

Long story short: MS saw an order of magnitude speed up in their STL implementation for float to string, allowing to make it an io bottleneck instead of a cpu bottleneck for writing float to io. They upstreamed it to clang libc++. We use float to string conversion a lot (JSON, CSV, SVG,...) in the ecosystem, so this kind of speedup matters.

## Milestones

- Open PR for replacing `io_lib_format:fwrite_g/1` with a ryu based implementation
- Work on merging or getting refused the PR
- Propose an EEP to bring `io_lib_format:fwrite_g/1` into `erlang:float_to_list/2` and `erlang:float_to_binary/2` BIF.
- Propose an EEP to change `erlang:float_to_list/2` and `erlang:float_to_binary/2` to use Ryu instead of glibc `sprintf`

## Reports & Dissemination

- Pull Request for `io_lib_format:fwrite_g/1` and work with the OTP team to merge it.
- EEP for further project
- Blog post/Conference talk are planned to talk about the work done, mentioning the support by the EEF.